

Drug -> Target -> Pathway -> Transcription -> Synergy

A biological guide for extending drug-synergy research

Core idea: A drug usually does not directly “affect a gene.” Most often, it binds a protein, changes that protein’s activity, propagates through pathways or networks, changes transcription factor activity, and only then changes gene expression. Your ML features should respect these layers.

Guide map

#	Topic	Why it matters
1	Central dogma	How DNA becomes RNA and protein.
2	Expression vs activity	Why mRNA, protein abundance, and activity are not equivalent.
3	Drug-target biology	How drugs bind proteins and change function.
4	Signaling pathways	How receptor signals become transcriptional programs.
5	Transcription factors and regulons	How pathway effects become gene-expression signatures.
6	Database edge types	Why DrugBank, STRING, Reactome, DoRothEA, and LINCS answer different questions.
7	Pathway and regulon scoring	How to compress thousands of genes into interpretable activity scores.
8	Network biology	How to use graphs, neighborhoods, propagation, and proximity.
9	Perturbational transcriptomics	How LINCS/L1000 directly measures drug-induced expression changes.
10	ML architecture implications	How to turn the biology into synergy-prediction features.

Reading rule: Whenever you see an arrow, ask what kind of relationship it represents: binding, regulation, phosphorylation, pathway membership, co-complex association, or observed expression response. These are not interchangeable.

1. The central dogma: DNA -> RNA -> protein

The central dogma is the basic information flow of molecular biology: DNA stores genetic instructions, RNA carries or helps process those instructions, and proteins do most of the functional work in the cell.



Figure 1. The basic information flow from genetic storage to cellular function.

A useful analogy is a restaurant: DNA is the master cookbook kept safely in the office. mRNA is a temporary photocopy of one recipe. The ribosome is the chef reading the copy. The protein is the finished dish, and the cell phenotype is the effect of that dish on the restaurant service.

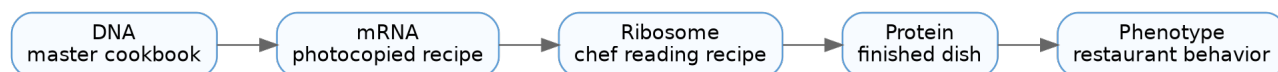


Figure 2. Restaurant analogy for gene expression.

Research consequence: RNA-seq mostly tells you about copied recipes, not necessarily how many machines were built or whether those machines are active.

Abbreviation	Expansion	Meaning in this section
DNA	Deoxyribonucleic acid	Stable genetic material.
RNA	Ribonucleic acid	A family of molecules involved in gene expression and regulation.
mRNA	Messenger RNA	The protein-coding RNA copy of a gene.

2. RNA, mRNA, abundance, and activity

RNA is a broad molecule family. mRNA is only one type of RNA: the type that carries protein-coding information. Other RNAs help build ribosomes, bring amino acids, or regulate expression.

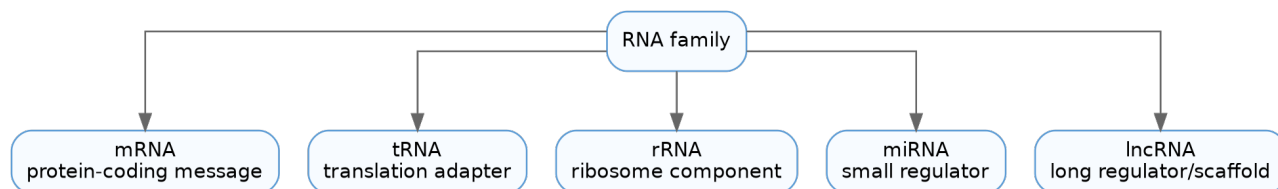


Figure 3. mRNA is a subtype of RNA, not a synonym for all RNA.

Layer	What it measures	Example experimental readout	What it does not prove
mRNA abundance	How much transcript exists for a gene.	RNA-seq, qPCR	Does not prove protein abundance or activity.
Protein abundance	How much protein exists.	Mass spectrometry, Western blot	Does not prove the protein is active.
Enzyme/protein activity	Whether the protein is functionally active.	Kinase assay, phospho-protein assay, substrate turnover	May change without mRNA change.

In a factory analogy, mRNA abundance is the number of printed work orders, protein abundance is the number of machines built, and enzyme activity is whether the machines are turned on, maintained, and actually producing output.

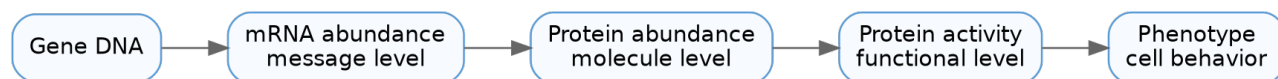


Figure 4. Expression, abundance, and activity are related but distinct layers.

Expression does not equal activity because cells regulate proteins after transcription. Translation efficiency, protein degradation, phosphorylation, localization, binding partners, inhibitors, and cleavage can all change activity without a proportional mRNA change.



Figure 5. Multiple regulatory gates sit between mRNA expression and activity.

Example: AKT mRNA may be high, but AKT signaling is usually better approximated by phosphorylated AKT and downstream pathway response, not by AKT transcript alone.

Abbreviation	Expansion	Meaning in this section
qPCR	Quantitative polymerase chain reaction	A method for measuring specific RNA or DNA amounts.
PTM	Post-translational modification	A chemical modification after a protein is made, such as phosphorylation.
AKT	Protein kinase B	A central kinase in the PI3K/AKT survival pathway.

3. How drugs relate to genes: usually indirectly

A drug usually begins with chemistry, not transcription. It binds one or more proteins, alters their activity, and the downstream cell response may eventually alter gene expression. Therefore, “drug affects gene” should usually be unpacked into several mechanistic steps.



Figure 6. The safer interpretation of drug-to-gene effects.

Concept	Definition	Research implication
Small molecule binding	A chemical compound physically interacts with a binding pocket on a protein.	Binding is the direct molecular event for many drugs.
Target protein	The protein the drug is intended or known to modulate.	Target annotations connect chemical space to biology.
Off-target effect	Binding to additional proteins besides the intended target.	Off-targets can cause side effects or unexpected synergy.
Mechanism of action	The chain of events by which a drug produces its biological effect.	MoA is more informative than chemical similarity alone.

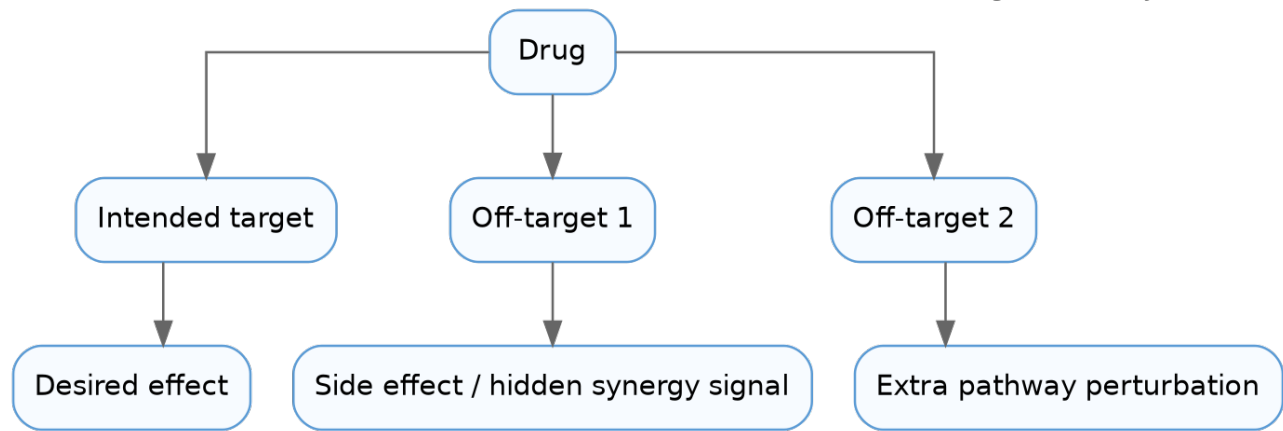


Figure 7. One compound can perturb multiple proteins.

IC50 and EC50 summarize dose-response behavior. IC50 is the concentration where inhibition reaches 50%. EC50 is the concentration where an effect reaches 50% of its maximal response. These values depend on assay conditions, cell line, time, endpoint, and target context.

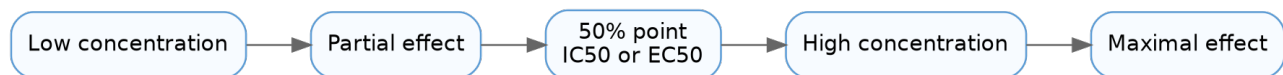


Figure 8. Conceptual dose-response progression.

Synergy intuition: Two drugs may be synergistic because they hit parallel survival mechanisms, different nodes in the same pathway, a target plus its resistance mechanism, or a direct target plus a hidden off-target dependency.

Abbreviation	Expansion	Meaning in this section
MoA	Mechanism of action	How a drug produces its biological effect.
IC50	Half-maximal inhibitory concentration	Dose causing 50% inhibition in a specific assay.
EC50	Half-maximal effective concentration	Dose causing 50% of maximal effect.

4. Signaling pathways: from receptor to transcriptional program

A signaling pathway is a communication chain. External or internal signals are detected, relayed through proteins, and often end by changing transcription factors and gene-expression programs.

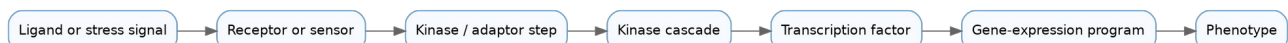


Figure 9. General signaling logic.

The company analogy: a customer complaint reaches reception, gets forwarded through managers, reaches an executive, and the executive changes company policy. Receptors are reception, kinase cascades are the forwarding managers, transcription factors are executives, and gene-expression changes are new employee instructions.

4.1 MAPK pathway

The MAPK pathway often promotes proliferation. It is one of the most important cancer signaling pathways because mutations or upstream receptor activation can keep growth signals on.



Figure 10. MAPK signaling from growth factor detection to proliferation genes.

Drug example: EGFR, BRAF, MEK, and ERK inhibitors target different levels of this same growth-signaling axis. For ML, a MAPK activity score may be more informative than one individual MAPK gene expression value.

Abbreviation	Expansion	Meaning in this section
MAPK	Mitogen-activated protein kinase	A signaling pathway family that relays growth/stress signals.
RTK	Receptor tyrosine kinase	A receptor that activates signaling through tyrosine phosphorylation.
RAS	Rat sarcoma small GTPase family	A molecular switch upstream of RAF.
RAF	Rapidly accelerated fibrosarcoma kinase family	A kinase downstream of RAS.
MEK	MAPK/ERK kinase	A kinase that activates ERK.
ERK	Extracellular signal-regulated kinase	A MAPK that changes downstream transcriptional programs.
TF	Transcription factor	A DNA-binding regulatory protein.

4.2 PI3K/AKT pathway

The PI3K/AKT pathway supports survival, metabolism, growth, and resistance. Cancer cells often use it to avoid death signals.

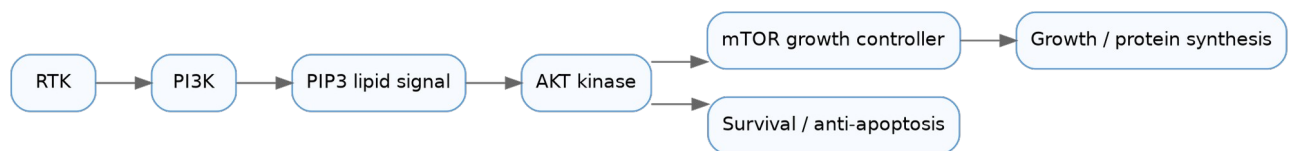


Figure 11. PI3K/AKT signaling supports survival and growth.

Synergy intuition: Combining a MAPK-pathway inhibitor with a PI3K/AKT-pathway inhibitor can block both proliferation and survival programs, which may be stronger than either alone in the right cell context.

Abbreviation	Expansion	Meaning in this section
PI3K	Phosphoinositide 3-kinase	An enzyme that creates lipid signals for AKT activation.
AKT	Protein kinase B	A survival and growth kinase downstream of PI3K.
PIP3	Phosphatidylinositol (3,4,5)-trisphosphate	A membrane lipid signal that helps recruit/activate AKT.
mTOR	Mechanistic target of rapamycin	A central growth and protein synthesis regulator.

4.3 JAK/STAT, p53, NF- κ B, cell cycle, apoptosis, and DNA damage response

Several other pathway systems are repeatedly important in drug response. JAK/STAT converts cytokine signals into transcriptional programs. p53 coordinates stress responses. NF- κ B coordinates inflammatory and survival programs. Cell-cycle machinery determines whether cells divide. Apoptosis determines whether cells execute programmed death. DNA damage response decides whether damage is repaired or escalated into arrest/death.

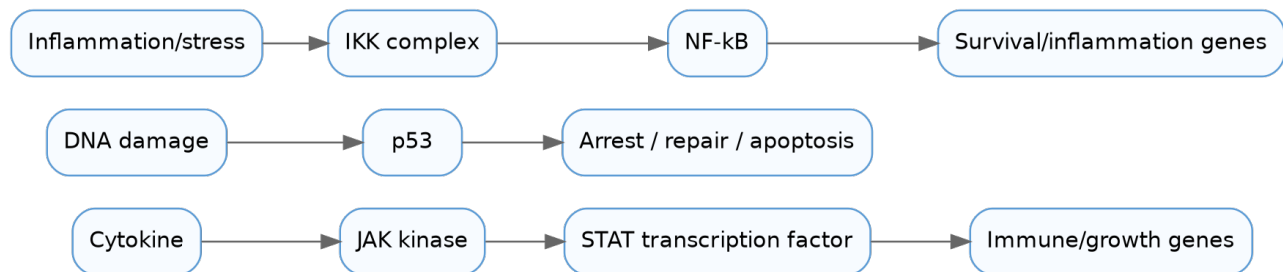


Figure 12. Three major signaling-response modules.

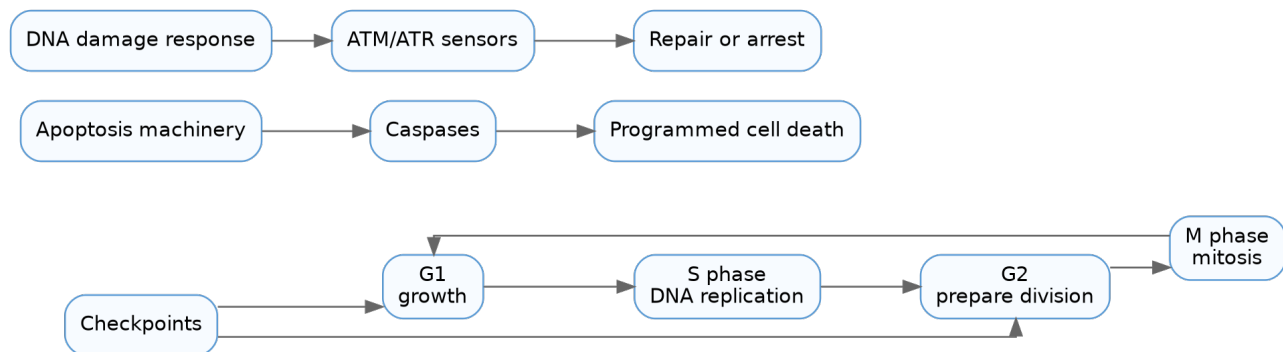


Figure 13. Cell-cycle, apoptosis, and DNA damage response modules.

Pathway/module	Usual biological role	Why it matters for drug response
JAK/STAT	Cytokine and immune/growth signaling.	Can drive proliferation, inflammation, immune evasion, and survival programs.
p53	Stress sensing, arrest, repair, apoptosis.	p53 mutation status can change response to DNA-damaging drugs.
NF-kB	Inflammation, stress, survival.	May protect cells from apoptosis and create resistance.
Cell cycle	Controls division phases and checkpoints.	CDK or mitotic inhibitors target this machinery.
Apoptosis	Programmed cell death.	Many synergies work by lowering the threshold for apoptosis.
DNA damage response	Senses and repairs DNA lesions.	PARP, ATR, ATM, CHK inhibitors exploit repair vulnerabilities.

Abbreviation	Expansion	Meaning in this section
JAK	Janus kinase	A kinase downstream of cytokine receptors.
STAT	Signal transducer and activator of transcription	A TF activated by JAK signaling.
p53	Tumor protein p53	A stress-response tumor suppressor TF.
NF-kB	Nuclear factor kappa-light-chain-enhancer of activated B cells	A TF family involved in inflammation and survival.
IKK	I-kappa-B kinase	A kinase complex that activates NF-kB signaling.
CDK	Cyclin-dependent kinase	A cell-cycle kinase family.
DDR	DNA damage response	The system that detects and responds to DNA damage.
ATM/ATR	Ataxia-telangiectasia mutated / ATM and Rad3-related	DNA damage sensor kinases.
PARP	Poly(ADP-ribose) polymerase	A DNA repair enzyme family.

5. Transcription factors and regulons

A transcription factor is a protein that binds regulatory DNA and changes transcription. A TF-target edge means “this TF regulates this gene.” A regulon is the full set of genes regulated by a TF, usually with information about activation or repression.

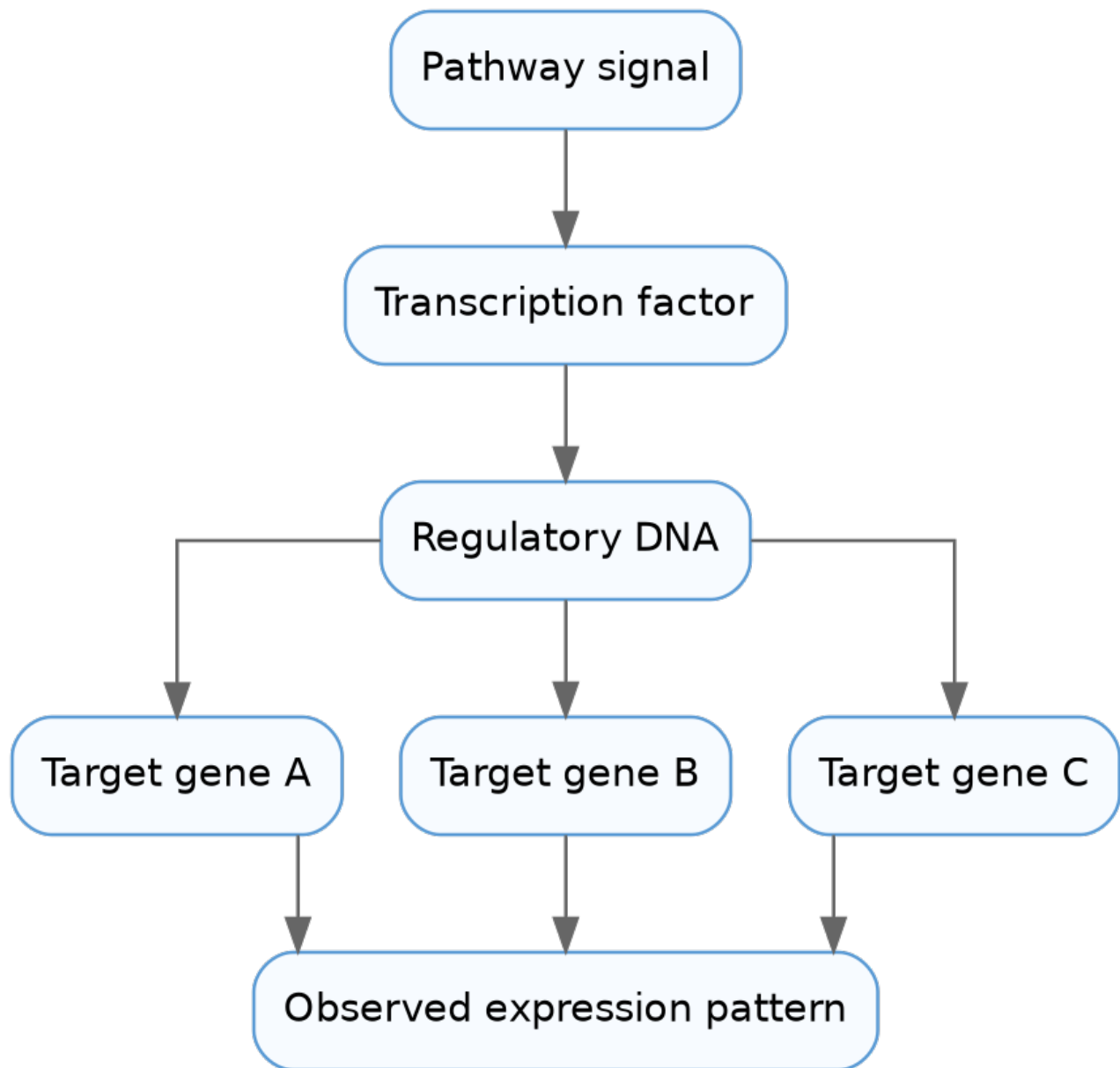


Figure 14. A pathway signal changes TF activity, and the TF changes many target genes.

TF activity is often inferred from its targets, not from the TF mRNA itself. If activated targets go up and repressed targets go down, that is evidence the TF became more active.



Figure 15. Regulon-based TF activity inference.

Analogy: you may not know whether the CEO is actively giving orders by measuring how often the CEO's name appears in emails. You infer CEO activity by checking whether many departments changed behavior in a coordinated direction.

Why this matters for your encoder: A regulon activity vector can transform noisy gene-level expression into a smaller set of biologically meaningful regulatory activities. This is often more interpretable than raw expression alone.

Abbreviation	Expansion	Meaning in this section
TF	Transcription factor	A DNA-binding regulator of gene transcription.
TF-target gene	Transcription factor target gene	A gene regulated by a specific TF.
Regulon	Regulatory module	The set of genes controlled by a regulator such as a TF.
DoRothEA	Discriminant Regulon Expression Analysis	A resource of signed TF-target interactions.
VIPER	Virtual Inference of Protein-activity by Enriched Regulon analysis	A method for inferring protein/TF activity from regulon expression patterns.

6. Pathway databases vs interaction databases

Biological databases are not interchangeable. Each database type describes a different edge meaning. Mixing them blindly creates noisy or biologically invalid features.

Database family	Examples	Typical edge meaning	Good use
Pathway/reaction databases	KEGG, Reactome	Molecule participates in a curated pathway, reaction, or event.	Map targets to known biological processes.
Protein interaction/association databases	STRING, BioGRID	Proteins interact physically or are functionally associated.	Build neighborhoods around targets.
Transcriptional regulation databases	DoRothEA, TRRUST, ChEA	TF regulates target gene.	Infer TF/regulon activity.
Drug-target databases	DrugBank, ChEMBL, DGIdb	Drug binds/modulates gene/protein target.	Connect chemical compounds to direct targets.
Perturbation/evidence databases	LINCS/L1000, CMap, CTD	Chemical perturbation changes expression or has chemical-gene evidence.	Use observed drug-response signatures.

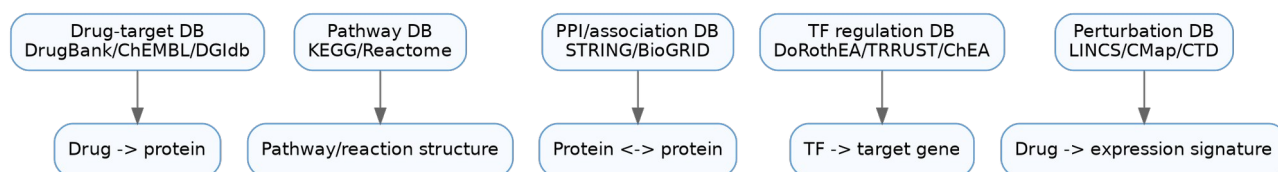


Figure 16. Different database families encode different biological relationships.

Practical rule: Use DrugBank/ChEMBL/DGIdb to get direct targets, STRING/Reactome/KEGG to expand biological context, DoRothEA/TRRUST/ChEA to interpret transcriptional regulation, and LINCS/L1000/CMap to observe actual drug-induced expression effects.

Abbreviation	Expansion	Meaning in this section
KEGG	Kyoto Encyclopedia of Genes and Genomes	A pathway and systems-biology database.
PPI	Protein-protein interaction	A physical or functional association between proteins.
STRING	Search Tool for the Retrieval of Interacting Genes/Proteins	A protein association network database.
BioGRID	Biological General Repository for Interaction Datasets	A database of genetic and protein interactions.
TRRUST	Transcriptional Regulatory Relationships Unraveled by Sentence-based Text mining	A TF-target regulation database.
ChEA	ChIP-X Enrichment Analysis	A resource based on chromatin immunoprecipitation evidence.
DGIdb	Drug-Gene Interaction database	A database for drug-gene interactions.
CTD	Comparative Toxicogenomics Database	Chemical-gene-disease evidence resource.
CMap	Connectivity Map	A perturbational gene-expression resource.

7. Gene set enrichment and pathway activity scoring

Raw expression can contain tens of thousands of genes. Gene set and pathway scoring methods compress this into interpretable biological programs such as MAPK activity, cell-cycle score, p53 activity, apoptosis score, or MYC regulon activity.



Figure 17. Compression from gene-level expression to biological program scores.

Method	Main idea	Typical output	Best use
GSEA	Tests whether a gene set is enriched near the top or bottom of a ranked differential-expression list.	Enrichment score per gene set.	Condition-vs-condition pathway interpretation.
ssGSEA	Single-sample version that scores gene sets within each sample.	Sample-by-gene-set matrix.	Cell-line feature generation.

GSEA	Estimates gene-set activity variation across samples.	Sample-by-pathway matrix.	Unsupervised pathway-level feature transformation.
PROGENy	Infers pathway activity from pathway-responsive footprint genes.	Pathway activity scores.	Footprint-based pathway inference.
VIPER	Infers TF/protein activity from downstream regulon expression.	Regulon/TF activity scores.	Regulatory activity inference.

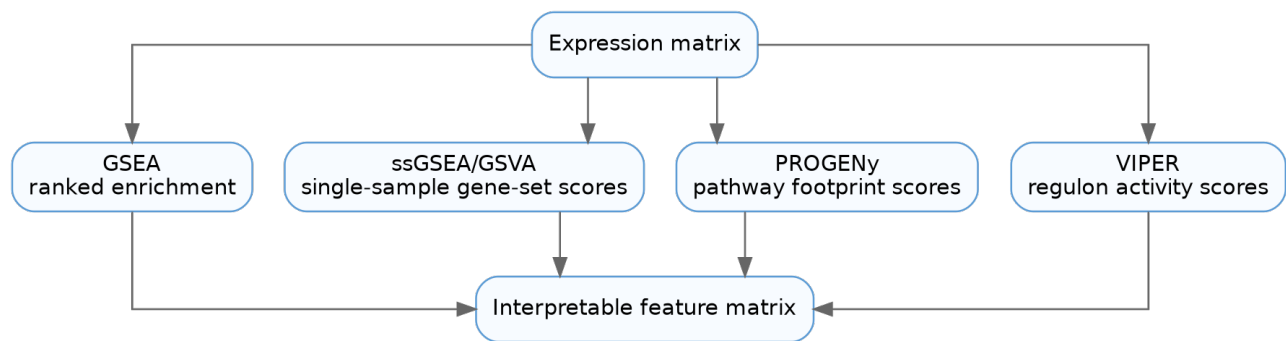


Figure 18. Several routes from expression to interpretable program scores.

Important distinction: A pathway member gene set asks whether pathway components are expressed. A pathway footprint method asks whether downstream response genes look like the pathway is active. These can differ substantially.

Abbreviation	Expansion	Meaning in this section
GSEA	Gene Set Enrichment Analysis	Tests enrichment of predefined gene sets in ranked data.
ssGSEA	Single-sample Gene Set Enrichment Analysis	Scores gene sets per individual sample.
GSVA	Gene Set Variation Analysis	Estimates pathway/gene-set activity variation across samples.
PROGENy	Pathway RespOnsive GENes for activity inference	Infers pathway activity from perturbation-derived footprint genes.
MYC	MYC proto-oncogene transcription factor	A major growth/proliferation-associated TF.

8. Network biology

Network biology represents biological systems as graphs. Nodes are entities such as proteins, genes, drugs, or pathways. Edges are relationships such as binding, regulation, interaction, or association. The edge type must always be explicit.

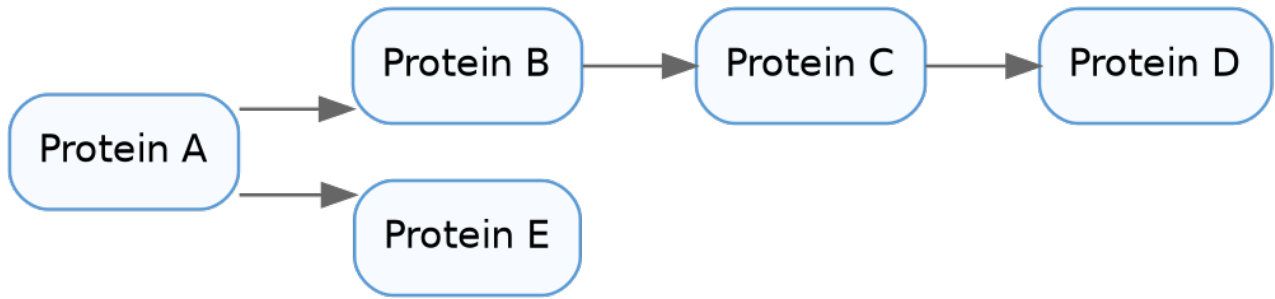


Figure 19. A simple protein network.

Network concept	Meaning	Drug-synergy use
Directed graph	Edges have direction, such as TF -> target gene.	Use for regulatory/signaling causality when known.
Undirected graph	Edges imply association without direction.	Use for PPI neighborhoods and proximity.
Graph neighborhood	Nodes close to a target.	Expand drug targets into affected modules.
Network propagation	Spread signal from seed targets across the graph.	Build smooth target-context features.
Random walk with restart	A walker moves through the graph but repeatedly jumps back to seed targets.	Score proteins by closeness to targets.
Network proximity	Distance between two target sets.	Compare whether drug pairs hit nearby or complementary modules.
Module detection	Find densely connected subgraphs.	Identify pathway-like biological communities.

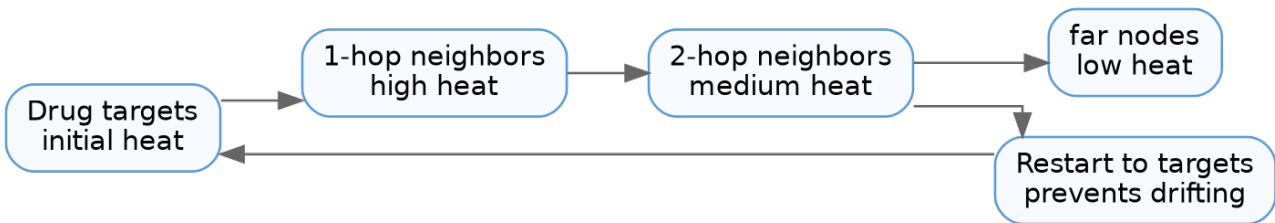


Figure 20. Network propagation / random walk with restart intuition.

Analogy: network propagation is like dropping ink into water. Proteins near the target become strongly colored, nearby regions become moderately colored, and distant regions remain mostly unchanged.

Synergy intuition: If two drugs hit the exact same target, they may be redundant. If they hit different nodes in the same pathway, they may reinforce the same biological blockade. If they hit parallel survival pathways, they may be strongly synergistic in a dependent cell line.

Abbreviation	Expansion	Meaning in this section
PPI	Protein-protein interaction	A network edge between proteins.
RWR	Random walk with restart	A network algorithm that repeatedly returns to seed nodes.
Seed node	Starting node	The initial target or gene set from which propagation begins.

9. Perturbational transcriptomics: the most direct drug-to-expression evidence

Perturbational transcriptomics measures gene expression after a perturbation. The perturbation can be a drug, gene knockout, knockdown, or overexpression. For the question “what expression program does this drug induce?”, this is one of the most direct data types.



Figure 21. Perturbational transcriptomics workflow.

Signature component	Meaning	Why it matters
Upregulated genes	Genes higher after treatment.	Shows activated response programs.
Downregulated genes	Genes lower after treatment.	Shows suppressed programs.
Dose	Drug concentration.	Low and high dose may induce different biology.
Time	Treatment duration.	Early signatures can be direct; late signatures can include secondary effects.
Cell line	Cellular context.	Mutations and basal state change the response.

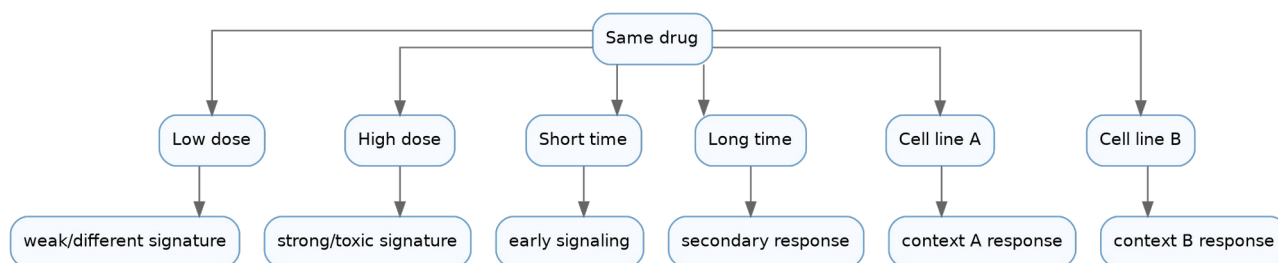


Figure 22. Drug signatures depend on dose, time, and cellular context.

Practical conclusion: For predicting drug effects on genes, do not rely only on KEGG or STRING. Use known targets and pathway context, but validate or augment with LINCS/L1000 perturbation signatures when available.

Abbreviation	Expansion	Meaning in this section
LINCS	Library of Integrated Network-based Cellular Signatures	A large perturbational profiling program.
L1000	Landmark 1000 gene expression assay	A reduced expression assay used by LINCS/CMap to infer broader transcriptional states.
RNA-seq	RNA sequencing	Sequencing-based measurement of RNA abundance.
KO	Knockout	Genetic removal/disruption of a gene.
KD	Knockdown	Reduction of gene expression, often by RNA interference.
OE	Overexpression	Artificially increasing expression of a gene.

10. Turning the biology into a synergy-prediction architecture

A strong mental model for your research has three layers: drug-target, network/pathway propagation, and transcriptional response. Your model can use each layer as a feature family rather than forcing raw SMILES and raw expression to learn everything alone.



Figure 23. Three-layer biological representation for synergy prediction.

A practical feature-building pipeline starts from known drug targets, maps them to proteins/genes, expands through curated pathways or protein networks, converts the affected biology into pathway or regulon scores, and compares those inferred features with observed LINCS/L1000 drug signatures.

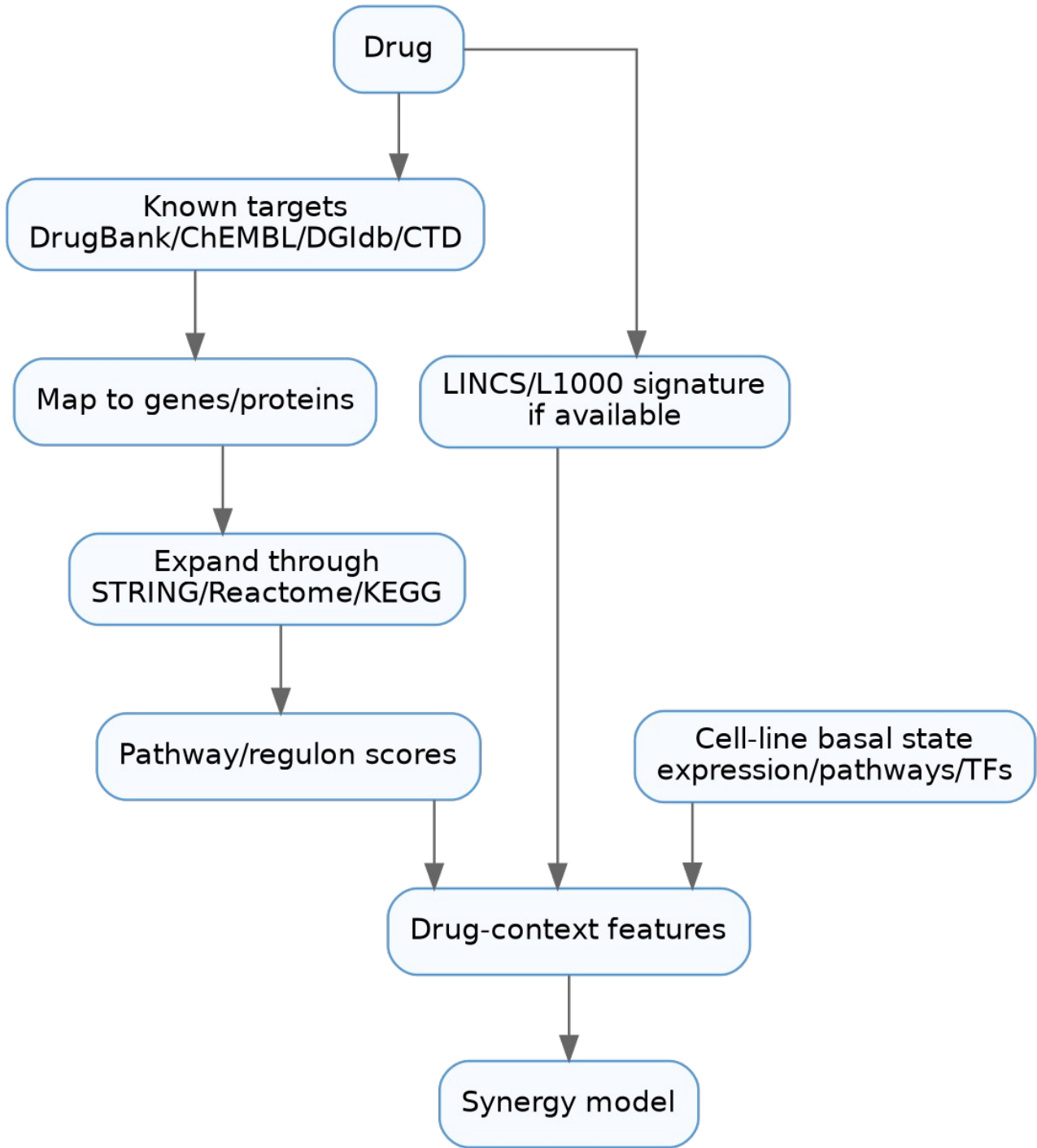


Figure 24. A biology-augmented drug-synergy feature pipeline.

Feature family	Examples	Main value
Chemical features	SMILES embeddings, molecular fingerprints.	Capture structure and chemical similarity.
Drug-target features	Known targets, target classes, target potency.	Capture direct mechanism.
Pathway/network features	Reactome/KEGG pathway membership, STRING neighborhoods, network	Capture downstream biological context.

	propagation scores.	
Perturbation features	LINCS/L1000 up/down signatures, drug-induced pathway scores.	Capture observed drug response.
Cell-line features	Basal expression, pathway activity, TF activity, mutations/CNV if available.	Capture context-specific vulnerabilities.
Pair features	Target overlap, pathway complementarity, network proximity, signature similarity/opposition.	Capture interaction between two drugs.

For a symmetric drug-pair model, Drug A + Drug B should produce the same prediction as Drug B + Drug A. A common design is to encode each drug separately and fuse them using sum, absolute difference, and elementwise product, together with the cell-line embedding.

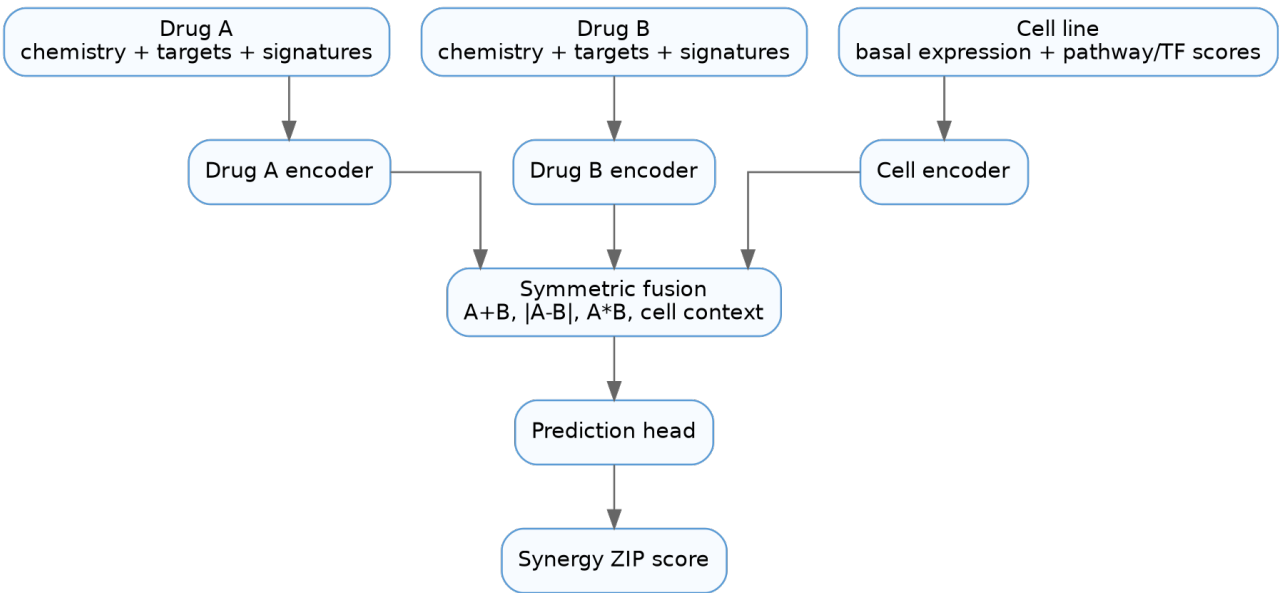


Figure 25. A biology-aware symmetric synergy architecture.

Recommended research extension: Start with a clear ablation ladder: baseline SMILES + expression; add known targets; add pathway scores; add regulon scores; add LINCS signatures; add network propagation. This tells you which biology actually helps.

Abbreviation	Expansion	Meaning in this section
SMILES	Simplified Molecular Input Line Entry System	A text representation of molecular structure.
CNV	Copy-number variation	Genomic gains or losses of DNA segments.
MLP	Multilayer perceptron	A feed-forward neural network.
ZIP	Zero Interaction Potency	A drug-synergy scoring model/metric.

11. Final synthesis

The best biological framing is not “drug affects gene” but “drug perturbs proteins, protein perturbations propagate through pathways and networks, transcription factors translate those pathway states into expression programs, and the cell-line context determines whether the final phenotype is survival, arrest, death, or synergy.”

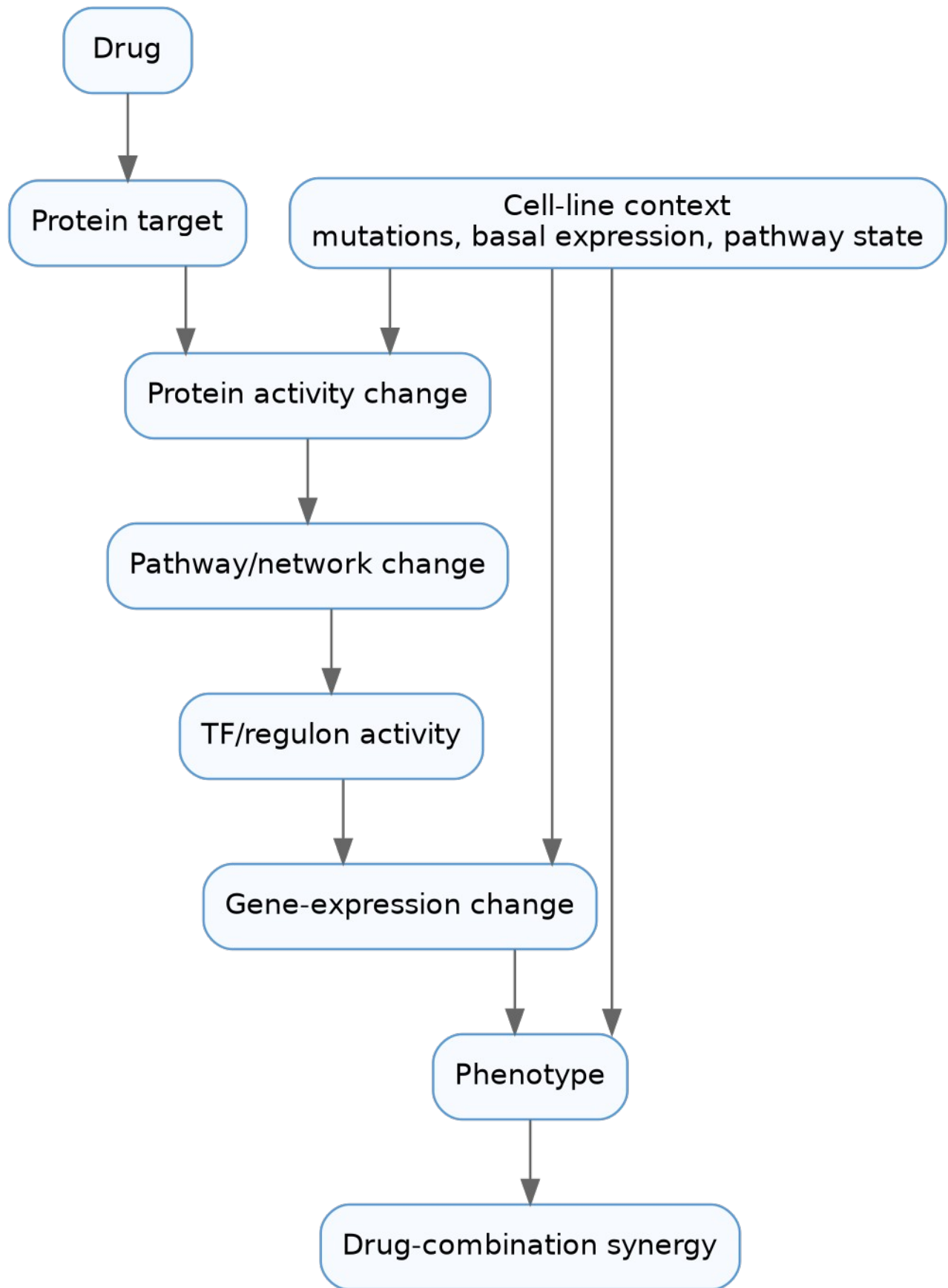


Figure 26. Integrated model of drug effect and synergy.

Take-home message: Use raw molecular data, but do not stop there. Transform drugs and cells into target, pathway, network, regulon, and perturbation-signature representations. Those features are more aligned with the actual biology of drug synergy.

Selected references and resources

- NCBI Bookshelf - Molecular biology / central dogma and gene expression resources.
- Reactome - curated pathway database and pathway browser.
- KEGG - pathway and systems biology maps.
- STRING - protein association network database.
- BioGRID - biological interaction datasets.
- DrugBank - drug and drug-target information.
- ChEMBL - bioactive molecules and target activity data.
- DGIdb - drug-gene interaction database.
- DoRothEA - signed transcription factor regulons.
- VIPER - regulon-based protein/transcription factor activity inference.
- PROGENy - pathway activity inference from perturbation-responsive genes.
- GSVA / ssGSEA / GSEA - gene set and pathway scoring frameworks.
- LINCS L1000 / Connectivity Map - perturbational transcriptomic signatures.
- CTD - chemical-gene-disease evidence.